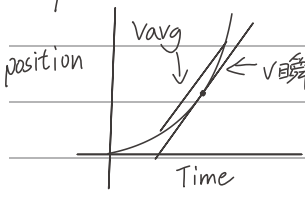
1. defination accelebration: the rate of change in velocity

Displacement: change in position $a=\frac{\Delta V}{\Delta t}$

Distance: length of route

Velocity: the rate of change in displacement

Speed: the rate of change in displacement



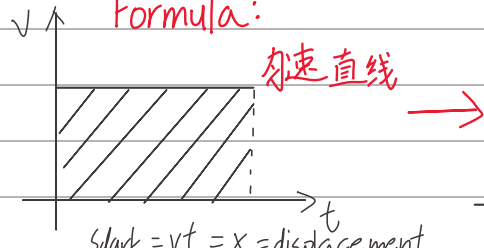
一个s-t图只能表示直线运动

5/20 ☆

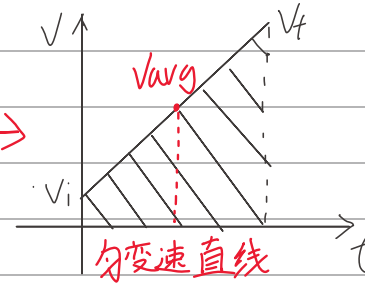
2 ☆ average accelebration = $\frac{\text{change in velocity}}{\text{time required for change}}$

$V_{avg}=\frac{V_i+V_f}{2}$ $X_{avg}=\frac{V_i+V_f}{2}t$ $V_f=V_i+at$

Formula:



$S_{dark}=vt=x=$ displacement

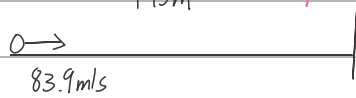


$S_{dark}=\frac{V_i+V_f}{2}t=$ displacement

$V_{avg}=\frac{V_i+V_f}{2}$	$V_f=V_i+at$	$X=\frac{V_i+V_f}{2}t$
cancel V_f		$X=V_it+\frac{1}{2}at^2$
V_i		$x=V_ft-\frac{1}{2}at^2$
t		$X=\frac{V_f^2-V_i^2}{2a}$

calculus

分割成无限多个小正方形



$195=-\frac{83.9^2}{2a}$ → $390a=-83.9^2$

$a=-\frac{83.9^2}{390}=18.0 m/s^2$

Practice:

3. Free Fall (不 G) $a=g=9.8m/s^2$

$v_i=0$

$v_f=at=gt$

$x=h=\frac{1}{2}at^2=\frac{1}{2}gt^2$

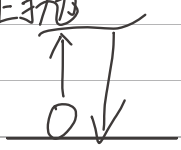
$x=h=\frac{v_f^2}{2a}=\frac{v_f^2}{2g}$ $v_f^2=2gh$.

(Assume no air resistance)

均根据匀变速直线运动公式推导

4. Projection $a=-g=9.8m/s^2$

1. 上抛



$v_f=v_i-gt$

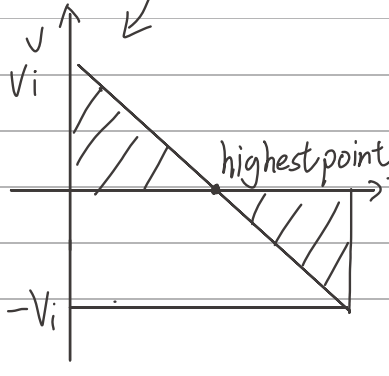
$x=h=v_it-\frac{1}{2}gt^2$

$x=h=\frac{v_i^2-v_f^2}{2g}$

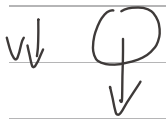
highest point (when $v_f=0$) $t=\frac{v_i}{g}$ $H=\frac{v_i^2}{2g}$

origin point ($H=0$) $t=\frac{2v_i}{g}$ $v_f=-v_i$

Why $2t_h=t_0$?



5. 下抛



$V_t = V_i + gt$

$h = V_i t + \frac{1}{2}gt^2$

$h = \frac{V_t^2 - V_i^2}{2g}$

5/23 practise

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① bearing:

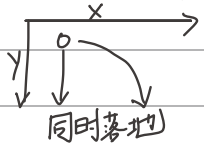
② N85.92°E

③ north of east

相对运动：向量减法

和速度：向量加法

5/31 平抛运动



$y: \begin{cases} v_y = gt \\ y = \frac{1}{2}gt^2 \end{cases}$

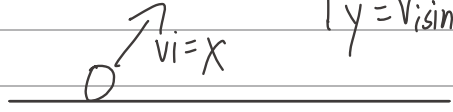
$x \begin{cases} v_x = v_i \\ x = v_i t \end{cases}$

6/4.

斜抛

$\begin{cases} v_x = v_i \cos\theta t \\ x = v_i \cos\theta t \end{cases}$

$\begin{cases} v_y = v_i \sin\theta - gt \\ y = v_i \sin\theta t - \frac{1}{2}gt^2 \end{cases}$



Range $\begin{cases} t = \frac{2v_i \sin\theta}{g} \\ X = v_i \cos\theta \, \frac{2v_i \sin\theta}{g} \\ x = \frac{v_i^2 \sin(2\theta)}{g} \end{cases}$

$\sin 2\theta = 2\cos\theta \sin\theta$

Force Dynamics

1. { Contact force
   Field forces
2. Newton's law

① Newton's first law (law of inertia)

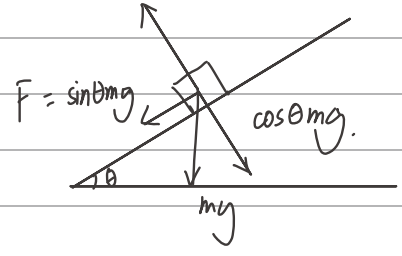
Tendency of an object not to accelebrate

Mass is the only measure of interia

② Newton's second law

F_{net} / $\Sigma F = ma$

The acceleration of an object is directly porportional to the net force acting on the object and inversely proportional to the subject's mass



$a = \frac{F}{m} = \sin\theta g$

③ Newton's second law

If two objects interact, the magnitude of the force exerted on object 1 by object 2 is equal to the magnitude of the force simultaneously

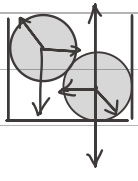
action and reaction
作用力 反作用力

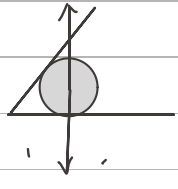
a. 大小相等. 方向相反. 同一直线 equal and opposite.
b. 不同物体 exerted on different objects
c. 同类型.
d. 同时有在, 同时消失

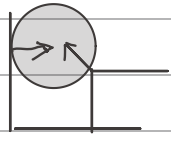
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从未遇到南北极 g↓.

1. 弹力







2 Hooke's law

$F=k\Delta x$ k (N/m)

5N ←〰→ 5N

5N ←〰→ 5N

受力相同

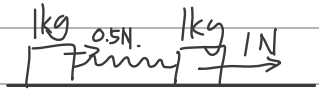
3. 连接体

先整体，后隔离



$$\begin{cases} F-(m_1+m_2)g=(m_1+m_2)a \\ m_2:\ kx-m_2g=m_2a \end{cases} \quad \frac{40}{1}-2g=2a \quad 10=kx \quad \& \ a=10$$

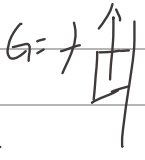
$$\begin{cases} a=10m/s \\ x=20cm \end{cases}$$

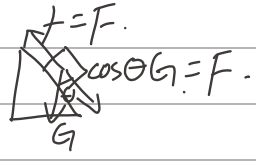


$$\begin{cases} F=(m_1+m_2)a \\ m_2:\ N=m_2a \end{cases}$$



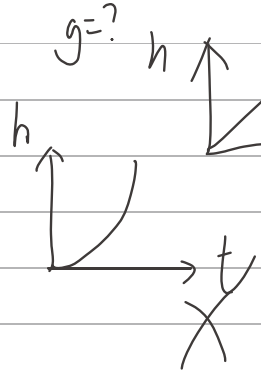
{static friction F_s
{kinetic friction F_k





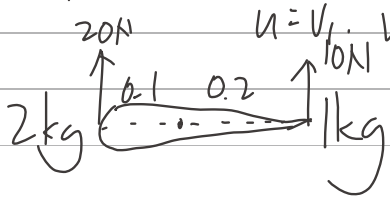


$f=\mu N \quad (\mu_s > \mu_k)$



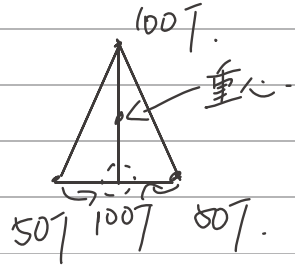
tree tall

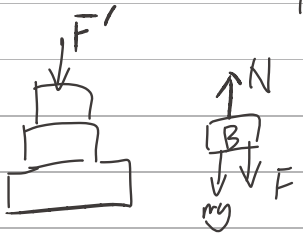
h	h_1	h_2	h_3
t	t_1	t_2	t_3
t^2	t_1^2	t_2^2	t_3^2



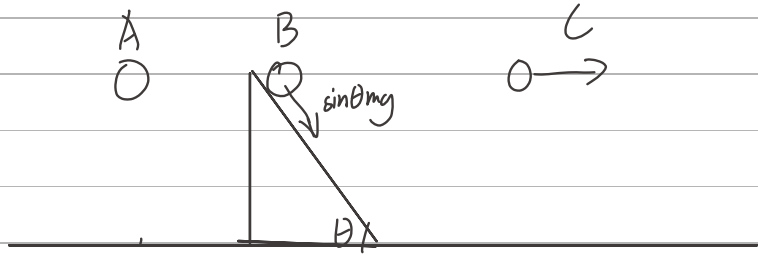
$u=v_1$. $v=v_4$.

$0.1\times 20 = 0.2\times 10$





$F=F'$



$t_B > t_A = t_C$

$v_A = v_B = v_C$

$\sin\theta mg = ma$.

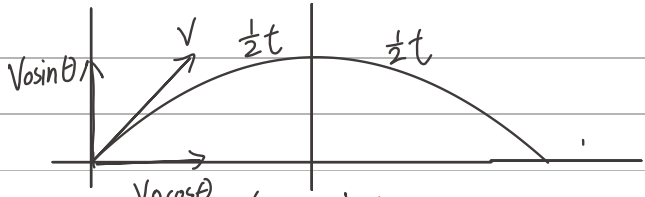
$s = vt + \frac{1}{2}at^2$

$\frac{h}{\sin\theta} = \frac{1}{2}g\sin\theta t_B^2$

$v_A = gt = g\sqrt{\frac{2h}{g}} = \sqrt{2gh}$

$v_B = at_B = g\sin\theta\sqrt{\frac{2h}{g\sin^2\theta}}$

$= \sqrt{2gh}$



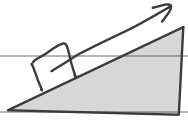
t: total time

$V_0\sin\theta = g\cdot\frac{1}{2}t$

h

$h_{max} = \frac{1}{2}g(\frac{1}{2}t)^2$

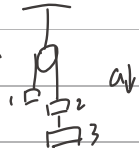
$V_0\sin\theta = g\cdot\frac{1}{2}t$



$a = g\sin\theta$

$m_2 + m_3 > m_1$

$m_2 < m_1$



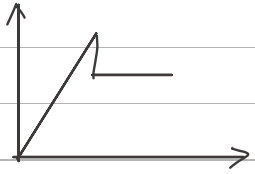
$m_3g - T_1 = m_3a$.

$m_2g + T_1 - T_2 = m_2a$.

$T_2 - m_1g = m_1a$

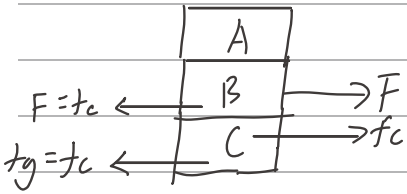
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1. fictional force.



$f_s \leq f_{smax}$.

$f_{smax} = \mu_s F_n$



$F - 3\mu_3 mg = 3ma$.

$\mu_3 = \frac{F - 3ma}{3mg}$,

$\mu_2 = \frac{F - 2ma}{2mg}$

